

# Exact Periodic Orbits and Algebraic Zeta Geometry of One-Vertex Edge-Type Dyck Shifts

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## Abstract

We close the periodic-point theory of the edge-type Dyck shift associated with one vertex and  $N$  loop edges. The algebraic zeta function yields explicit odd and even fixed-count formulas for every  $N$  and every period, followed by primitive points and orbits through Möbius inversion. We distinguish the  $N = 1$  full-two-shift boundary from the  $N > 1$  double dominant pole, quadratic branchpoints, nonrationality, and orbit asymptotics. Formal series, symbolic algebra, and direct periodic-word enumeration independently audit the result; the source-locked entropy is  $\log(N + 1)$ .

## 1 Model and algebraic zeta

Let  $D_N^E$  be the edge-type Dyck shift for the graph with one vertex and  $N$  loop edges. Its alphabet is  $\{a_i, b_i : 1 \leq i \leq N\}$ , with inverse-monoid relation  $a_i b_j = 1$  for  $i = j$  and 0 otherwise. A finite block is admissible when its reduction is nonzero. Put

$$s_N(z) = \sqrt{1 - 4Nz^2}.$$

The context-free circular-code series satisfies

$$g_N(z) = \frac{Nz^2}{1 - g_N(z)}, \quad g_N(z) = \frac{1 - s_N(z)}{2},$$

where the second expression is the small solution. The source theorem, specialized to the one-vertex  $N$ -loop graph, gives

$$\zeta_N(z) = \frac{1 - g_N(z)}{(1 - Nz - g_N(z))^2}.$$

Substitution, rather than a claim of priority, yields

$$\zeta_N(z) = \frac{2(1 + s_N(z))}{(1 + s_N(z) - 2Nz)^2}. \quad (1)$$

**Theorem 1** (all fixed and primitive counts). *Let  $F_N(n) = |\text{Fix}(\sigma^n)|$ . If  $n$  is odd, then*

$$F_N(n) = 2 \left( (N + 1)^n - \sum_{j=0}^{(n-1)/2} \binom{n}{j} N^j \right).$$

*If  $n$  is even, then*

$$F_N(n) = 2 \left( (N + 1)^n - \sum_{j=0}^{n/2} \binom{n}{j} N^j \right) + \binom{n}{n/2} N^{n/2}.$$

*The numbers of least-period points and primitive orbits are*

$$P_N(n) = \sum_{d|n} \mu(n/d) F_N(d), \quad C_N(n) = P_N(n)/n.$$

*Proof.* By definition,  $z\zeta'_N(z)/\zeta_N(z) = \sum_{n \geq 1} F_N(n)z^n$ . Insert (1), expand the quadratic series  $s_N(z)$ , and separate the central binomial term when  $n$  is even. The two displayed parity formulas result. Möbius inversion isolates least periods. A least-period- $n$  orbit has exactly  $n$  possible origins, proving the last division.  $\square$

## 2 The origin-marked word convention

$F_N(n)$  counts points of  $\text{Fix}(\sigma^n)$ , equivalently length- $n$  periodic words with a distinguished origin; it does not count cyclic necklaces. Our direct audit enumerates such words and checks every cyclic factor through length  $2n$  in the periodic extension. A stack reduction cancels  $a_i b_i$  and rejects  $a_i b_j$  for  $i \neq j$ . Thirty-three  $(N, n)$  cells, including  $N = 1, \dots, 6$ , agree exactly with Theorem 1. Only after Möbius inversion is  $P_N(n)$  divided by  $n$ .

This distinction is substantive: division of  $F_N(n)$  before inversion would mix points whose least period properly divides  $n$  and generally fail integrality.

**Lemma 1** (two-period sufficiency). *An  $n$ -periodic extension is admissible if and only if every cyclic factor of length at most  $2n$  has nonzero reduction.*

*Proof.* Every nonzero reduced word has normal form  $BA$ , with unmatched closing letters  $B$  followed by unmatched opening letters  $A$ . Fix a cyclic origin and call its period word  $v$ . If  $v$  reduces to zero, a forbidden prefix already has length at most  $n$ . Otherwise, in  $v^2$  the only new interaction is the  $AB$  interface. A colour mismatch is therefore already visible there. If the interface is compatible, equal lengths cancel and unequal lengths leave a monotone excess on one side; crucially, the final copy of  $A$  is unchanged. Every later copy, and every prefix of it, repeats the same interface comparison. Thus the first zero prefix has length at most  $2n$ . Auditing every cyclic origin covers every factor of the bi-infinite extension.  $\square$

## Source note

The model, zeta formula, and entropy lock follow Krieger–Matsumoto [1]; Keller’s loop-counting framework gives related context [2]. The official Münster volume PDF controls pagination; an arXiv journal-reference entry reports 171–185. No priority claim is made.

## References

- [1] W. Krieger and K. Matsumoto, Zeta functions and topological entropy of the Markov-Dyck shifts, *Münster J. Math.* 4 (2011), 171–184, arXiv:0706.3262.
- [2] G. Keller, Circular codes, loop counting, and zeta-functions, *J. Combin. Theory Ser. A* 56 (1991), 75–83, doi:10.1016/0097-3165(91)90023-A.

## Declarations

**Scope.** The exact registered literal is NO\_BAD\_EULER\_OR\_ROOT\_NUMBER. **Data and code.** Exact ledgers, independent checks, and build instructions are supplied in the HCS-C205 release package. **Competing interests.** None declared. **AI-use disclosure.** Generative AI tools assisted drafting and code generation. Every theorem statement, source record, exact output, and scope claim was checked through the released internal artifact audit chain; this is not external peer review or an independent error process.